Harshith Reddy Nalla

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Summary

AI Engineer specializing in production deep learning systems for medical imaging. Published at **AAAI 2026**, delivering PyTorch/TensorFlow models achieving **92%+ accuracy** in clinical diagnostics with full-stack deployment (FastAPI, Docker, AWS).

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL
AI/ML Stack: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, OpenCV, Keras
Deep Learning: CNNs, Transformers, Mamba Architectures, Computer Vision, Medical Image Segmentation, Transfer Learning
Backend/APIs: FastAPI, Spring Boot, RESTful APIs, Spring Framework
Frontend: React, TypeScript, Tailwind CSS, Responsive Design
MLOps/Cloud: Docker, Kubernetes, AWS (EC2, S3, ECS, EKS, Pipeline), Hugging Face Spaces, CI/CD
Data & Tools: NumPy, Pandas, SQL Server, PostgreSQL, MySQL, Git, Model Evaluation (Dice, IoU, F1, ROC-AUC)

Experience

AI Research Assistant June 2025 – Present
University of South Dakota, Computer Science Department *Vermillion, SD*

- Developed end-to-end deep learning pipelines using **PyTorch CNNs, Transformers, and Mamba**, achieving **92% accuracy** in pancreatic cancer detection with **25% improvement** over baselines through advanced preprocessing and hyperparameter tuning.
- Built production full-stack AI systems with **React, Spring Boot, FastAPI, and Docker**, enabling real-time inference on **500+ scans** (sub-2s latency), 3D visualization, and automated reports reducing radiologist time.
- Engineered ML evaluation framework with comprehensive metrics (Dice, IoU, sensitivity, specificity), conducted ablation studies across **1,000+ cases**, contributing to **AAAI 2026 publication** on CNN architectural advantages.
- Collaborated with clinicians to translate requirements into technical specs, refined models via domain feedback, and deployed AI tools improving cancer detection in real healthcare workflows.

Entrepreneurial Lead, NSF I-Corps Program October 2025 – November 2025
National Science Foundation – DentiQuest AI *Vermillion, SD*

- Led customer discovery for AI dental diagnostics, conducting **50+ stakeholder interviews** with dentists and clinicians to validate product-market fit and refine model requirements for commercial deployment.

Projects

Liver Profile AI | *PyTorch, 3D CNNs, Mamba, FastAPI, Docker, Hugging Face, React* 2025

- Developed automated liver segmentation using **3D CNNs and Mamba architectures**, achieving **88% Dice score** on LiTS dataset and **95% accuracy** across 200+ scans.
- Deployed **FastAPI microservice** with real-time 3D visualization to **Hugging Face Spaces**, implementing efficient inference pipeline with clinical interpretation overlays.

DentiMap | *PyTorch, CNNs, Spring Boot, React, FastAPI, TypeScript* 2025

- Built full-stack dental diagnostics app with **PyTorch CNNs** achieving **87% accuracy** in caries detection with sub-1-second inference.
- Deployed **FastAPI microservices backend** with AI model on Hugging Face Spaces and integrated a React/TypeScript frontend via RESTful APIs for real-time inference and seamless model serving.

Publications

When CNNs Outperform Transformers and Mambas – AAAI 2026 Workshop Nov 2025
∗ A. Ghimire, J. Zeng, R. Paudel, N. K. Tomar, D. R. Nayak, **H. R. Nalla**, V. Jha, G. Reynolds, D. Jha

Education

University of South Dakota Expected May 2027
Bachelor of Science in Computer Science, AI Concentration | GPA: 3.6/4.0 | DSA Certificate- Java *Vermillion, SD*